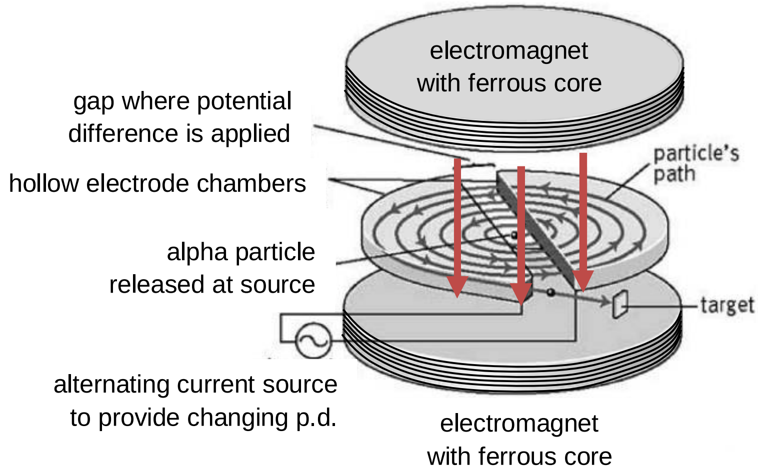
4	**(a)**		region of space where a magnetic pole (or magnet), moving charged particle or current-carrying conductor will experience a magnetic force	
	(b)			

	(c)	**(i)**	Magnetic force provides for centripetal force for particles to move in a circular path: $\frac{mv^2}{r} = qvB$ $r = \frac{mv}{qB}$ Mass of alpha particle = 4u Charge of alpha particle = +2e $$r = \frac{4\text{u}v}{2\text{e}B} = \frac{2\text{u}v}{\text{e}B}$$ Correct substitution of both mass and charge in terms of u and e	
		(ii)	Only dependent variable is velocity as *B*, u and e are constant. Each time ions pass between, they are <u>accelerated by the electric field</u> due to potential difference between the dees. OR <u>gains kinetic energy</u> from potential difference (from electric potential energy) The <u>velocity increases, therefore radius increases</u>.	
	(d)		Magnetic flux density at centre of solenoid is given by $B = \mu_0 nI$ $B = (4\pi \times 10^{-7})\left(\frac{3100}{0.39}\right)(200)$ $B = 1.998$ T = 2.0 T (shown)	

	(e)		Magnetic force provides for centripetal force for particles to move in a circular path: $mv\omega = qvB$ $\omega = \dfrac{qB}{m}$ $T = \dfrac{2\pi m}{qB}$ $T = \dfrac{2\pi(4)(1.66 \times 10^{-27})}{2(1.6 \times 10^{-19})(2.0)}$ $T = 6.519 \times 10^{-8}$ s Every complete circle/cycle, the particle is accelerated twice Total number of times accelerated $= 0.52\times10^{-3} / 6.519\times10^{-8} \times 2$ = 15954 times Energy gained per acceleration across dees $= \frac{(16\times10^{6})(1.6\times10^{-19})}{15954}$ $= 1.6047\times10^{-16}$ J Energy gained = charge x potential difference p.d. = $\frac{1.6047\times10^{-16}}{(2)(1.6\times10^{-19})}$ = 501.5 V = 500 V (2 sf)	
5	(a)	(i)	Work function is defined as the minimum energy required to eject an	